

Ancestral antibiotic resistance in *Mycobacterium tuberculosis*



Chemotherapeutic options to treat tuberculosis are severely restricted by the intrinsic resistance of *Mycobacterium tuberculosis* to the majority of clinically applied antibiotics. Such resistance is partially provided by the low permeability of their unique cell envelope. Here we describe a complementary system that coordinates resistance to drugs that have penetrated the envelope, allowing mycobacteria to tolerate diverse classes of antibiotics that inhibit cytoplasmic targets. This system depends on *whiB7*, a gene that pathogenic *Mycobacterium* shares with *Streptomyces*, a phylogenetically related genus known as the source of diverse antibiotics. In *M. tuberculosis*, *whiB7* is induced by subinhibitory concentrations of antibiotics (erythromycin, tetracycline, and streptomycin) and *whiB7* null mutants (*Streptomyces* and *Mycobacterium*) are hypersusceptible to antibiotics in vitro. *M. tuberculosis* is also antibiotic sensitive within a monocyte model system. In addition to antibiotics, *whiB7* is induced by exposure to fatty acids that pathogenic *Mycobacterium* species may accumulate internally or encounter within eukaryotic hosts during infection. Gene expression profiling analyses demonstrate that *whiB7* transcription determines drug resistance by activating expression of a regulon including genes involved in ribosomal protection and antibiotic efflux. Components of the *whiB7* system may serve as attractive targets for the identification of inhibitors that render *M. tuberculosis* or multidrug-resistant derivatives more antibiotic-sensitive.